



AMERICAN INTERNATIONAL UNIVERSITY – BANGLADESH (AIUB)

Faculty of Science and Technology (FST)

Department of Computer Science

Course Name: Compiler Design

Section: [E]

Semester: Fall 2021-2022

Submission Date: 10.12.2021

Term: Final-Term

Faculty Name: Md Masum Billah

1. Check the grammar below and justify whether the grammar is LL(1) or not:

$S \rightarrow U$

$U \rightarrow T a U$

$U \rightarrow T a T$

$T \rightarrow a T b T$

$T \rightarrow b T a T$

$T \rightarrow d$

2. Which of the following grammars are ambiguous? Show **LL(1) parsing table** using **First** and **Follow** to determine whether the grammar is ambiguous or not.

1) $S \rightarrow 0A \mid 1B$

$A \rightarrow 0AA \mid 1S \mid 1$

$B \rightarrow 1BB \mid 0S \mid 0$

2) $S \rightarrow aSb \mid bSa \mid SS \mid \epsilon$

3) $S \rightarrow 1S1 \mid T$

$T \rightarrow 1X1 \mid X$

$X \rightarrow 0X0 \mid 1$

3. Design a context free grammar by your own. Construct a **LL(1) parsing table** using **First** and **Follow** and show **stack movement** for any one input string.

Instructions For Preparing Your Assignment:

- Your assignment must be in the form of **handwritten hard copy**, not the typing copy.
- You can use a **scanner to create a PDF file** of your completed handwritten assignment, or you can **use your smartphone to take images of your work that can be compiled in a Word document and convert it into PDF**. Please check the video tutorial who cannot understand how to make a PDF.
 - ~ Scanner: (<https://www.youtube.com/watch?v=fPp8Ox7qfCM>)
 - ~ Images: (<https://www.youtube.com/watch?v=5K7xxQ8np90>)
- Rename the PDF as your student ID_CD_Asg_Final_Sec **(Example: 19-12345-2_CD_Asg_Final_Term.pdf)**
- **Copy and paste is prohibited**. I will check everyone's assignment carefully.
- Submit the assignment on 10 December 2021 (Friday) 09:59 PM. **Deadline will be strictly followed** and marks will be deducted who misses submission deadline.
- You must **add a cover page**. The key contents of your cover page should be (Course Title, Submission Type [e.g. Final-Assignment], Your Name, ID, Section, Department and Date of Submission).

Instructions For Submitting Your Assignment:

- 1) Complete all the tasks and create a PDF of your assignment.
- 2) Rename the PDF as your student ID_CD_Asg_Final_Sec (Example:19-12345-2_CD_Asg_Final__.pdf)
- 3) Attach the file here (**Add Work**).
- 4) **Turn in / Hand in.**
- 5) **Submit the assignment on 10 December 2021 (Friday) 09:59 PM.**
- 6) For Submission Instruction, please check the video tutorial (<https://www.youtube.com/watch?v=hgJAUupKkfM>)